# Data Communication & Computer Networks (CN)

## Topic 7 of 7 : Network Security

**44 MCQs** — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **44** MCQs + **22** Nukte rapid-fire revision points

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

## Syllabus Coverage — sub-topic wise order


**1.** Section A — Security Goals & Attacks — Q1 - Q15

**2.** Section B — Cryptography — Q16 - Q34

**3.** Section C — Firewalls, IPSec & Security Protocols — Q35 - Q44


**How to read:** Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

**System:** generated by **tools/cn_build_topic.py** from **tools/data/topic7_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

## Section A — Security Goals & Attacks

CIA triad, attack-goal mapping (interception/modification/DoS/spoofing), active vs passive attacks, malware phases, security tools.

**Q1.** The term 'CIA triad' in information security refers to:

Information security में 'CIA triad' शब्द किसे कहते हैं?

**(A)** Cryptography, Integrity, Authentication
**(B)** Confidentiality, Integrity, Availability
**(C)** Confidentiality, Information, Access Control **(D)** Information, Access

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Confidentiality, Integrity, Availability** — The CIA triad = Confidentiality (only authorised access), Integrity (no unauthorised change), Availability (accessible when needed) — the three core security goals.

CIA triad = Confidentiality (केवल authorised access), Integrity (बिना authorisation बदलाव नहीं), Availability (ज़रूरत पर उपलब्ध) — तीन मूल security goals।

★ **Nukta:** CIA = Confidentiality, Integrity, Availability. AAA (authentication, authorisation, accounting) is a separate trio.

CIA = Confidentiality, Integrity, Availability। AAA (authentication, authorisation, accounting) अलग trio है।

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**Q2.** The goal of 'Integrity' ensures that:

'Integrity' का लक्ष्य सुनिश्चित करता है कि:

**(A)** Data can be accessed only by authorised users **(B)** Data is accurate and has not been modified
**(C)** Data is always available when needed **(D)** Unauthorised users cannot enter the system

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Data is accurate and has not been modified** — Integrity means data remains accurate and unaltered except by authorised people — checksums/digital signatures protect it.

Integrity का अर्थ है data सही और अपरिवर्तित रहे (केवल authorised लोग ही बदलें) — इसे checksums/digital signatures सुरक्षित रखते हैं।

★ **Nukta:** Goal map: Confidentiality = authorised access only. Integrity = no unauthorised change. Availability = always accessible.

Goal map: Confidentiality = केवल authorised access। Integrity = बिना authorisation बदलाव नहीं। Availability = हमेशा उपलब्ध।

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**Q3.** Which of the following is NOT a goal of network security?

निम्न में से कौन-सा network security का लक्ष्य नहीं है?

**(A)** Confidentiality **(B)** Integrity
**(C)** Availability **(D)** Redundancy

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides]

✔ **Correct: (D) Redundancy** — Redundancy (duplicate links/equipment) improves fault tolerance — it is a reliability technique, not a security goal.

Redundancy (duplicate links/equipment) fault tolerance बढ़ाती है — यह reliability तकनीक है, security लक्ष्य नहीं।

★ **Nukta:** Security goals = CIA. Reliability goals = redundancy, fault tolerance, recovery.

Security goals = CIA। Reliability goals = redundancy, fault tolerance, recovery।

**Q4.** A modification attack violates which security goal?

Modification attack कौन-से security लक्ष्य का उल्लंघन करता है?

**(A)** Confidentiality **(B)** Integrity
**(C)** Availability **(D)** Authentication

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Integrity** — A modification attack changes data in transit — that breaks INTEGRITY (data is no longer accurate/unaltered).

Modification attack रास्ते में data बदल देता है — यह INTEGRITY तोड़ता है (data अब सही/अपरिवर्तित नहीं)।

★ **Nukta:** Attack→goal: Eavesdropping/Interception → Confidentiality. Modification → Integrity. DoS/Interruption → Availability. Spoofing → Authentication.

Attack→goal: Eavesdropping/Interception → Confidentiality। Modification → Integrity। DoS/Interruption → Availability। Spoofing → Authentication।

**Q5.** Which of the following attacks violates confidentiality?

निम्न में से कौन-सा attack confidentiality का उल्लंघन करता है?

**(A)** Interception **(B)** Modification
**(C)** Fabrication **(D)** Interruption

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides]

✔ **Correct: (A) Interception** — Interception (eavesdropping) lets an unauthorised party READ the data — breaking confidentiality.

Interception (eavesdropping) से unauthorised पक्ष data पढ़ लेता है — confidentiality टूटती है।

★ **Nukta:** Attack map: Interception→Confidentiality, Modification→Integrity, Fabrication→Authenticity, Interruption→Availability.

Attack map: Interception→Confidentiality, Modification→Integrity, Fabrication→Authenticity, Interruption→Availability।

**Q6.** A Denial of Service (DoS) attack violates which goal?

Denial of Service (DoS) attack कौन-से लक्ष्य का उल्लंघन करता है?

**(A)** Confidentiality **(B)** Integrity
**(C)** Availability **(D)** Authentication

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (C) Availability** — A DoS attack floods a server/network so legitimate users cannot access the service — breaking AVAILABILITY.

DoS attack server/network को flood करके legitimate users को service से रोकता है — Availability टूटती है।

★ **Nukta:** DoS = Availability attack. DDoS = distributed DoS. Smurf/ SYN-flood = DoS variants.

DoS = Availability attack। DDoS = distributed DoS। Smurf/ SYN-flood = DoS variants।

**Q7.** A spoofing attack targets which security aspect?

Spoofing attack किस security पहलू को लक्षित करता है?

**(A)** Authentication **(B)** Integrity
**(C)** Availability **(D)** Confidentiality

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (A) Authentication** — Spoofing pretends to be someone else (fake IP/MAC/email id) — it attacks AUTHENTICATION (who you claim to be).

Spoofing किसी और का रूप धारण करता है (fake IP/MAC/email id) — यह AUTHENTICATION पर हमला है (आप कौन हैं)।

★ **Nukta:** Spoofing/impersonation → Authentication. Phishing → steals credentials (auth). Sniffing → Confidentiality.

Spoofing/impersonation → Authentication। Phishing → credentials चुराता है (auth)। Sniffing → Confidentiality।

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**Q8.** Which of the following is an ACTIVE attack?

निम्न में से कौन-सा ACTIVE attack है?

**(A)** Interception **(B)** Modification
**(C)** Traffic analysis **(D)** Eavesdropping

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Modification** — Modification is an active attack — the attacker CHANGES the data stream; interception/traffic analysis/eavesdropping are passive (they only observe).

Modification एक active attack है — attacker data stream बदलता है; interception/traffic analysis/eavesdropping passive हैं (केवल देखते हैं)।

★ **Nukta:** Passive = observe only (eavesdropping, traffic analysis). Active = change/disrupt (modification, fabrication, interruption/DoS).

Passive = केवल देखना (eavesdropping, traffic analysis)। Active = बदलना/बाधा (modification, fabrication, interruption/DoS)।

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**Q9.** The primary difference between active and passive attacks is:

Active और passive attacks के बीच मुख्य अंतर क्या है?

**(A)** Active attacks change data; passive attacks do not **(B)** Passive attacks destroy hardware
**(C)** Both are the same **(D)** Passive attacks are legal

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (A) Active attacks change data; passive attacks do not** — Active attacks intercept AND modify/disrupt the communication; passive attacks only eavesdrop without changing anything.

Active attacks संचार को intercept करके बदलते/बाधित करते हैं; passive attacks केवल सुनते हैं, कुछ नहीं बदलते।

★ **Nukta:** Active: modification, fabrication, interruption. Passive: eavesdropping, traffic analysis. Exam one-liner: 'change vs observe'.

Active: modification, fabrication, interruption। Passive: eavesdropping, traffic analysis। Exam one-liner: 'change बनाम observe'।

**Q10.** In the virus life cycle, which phase actually performs the malicious activity?

Virus life cycle में कौन-सा phase वास्तव में malicious activity करता है?

**(A)** Infection Phase **(B)** Triggering Phase
**(C)** Dormant Phase **(D)** Execution (Payload) Phase

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (D) Execution (Payload) Phase** — The execution phase runs the payload — the actual damage (deleting files, corrupting data).

Execution phase payload चलाता है — वास्तविक नुकसान (files delete करना, data खराब करना)।

★ **Nukta:** Virus phases: Dormant (idle) → Propagation (copy) → Triggering (condition met) → Execution (payload runs).

Virus phases: Dormant (idle) → Propagation (copy) → Triggering (condition met) → Execution (payload runs)।

**Q11.** Which backup medium provides Write-Once-Read-Many (WORM) functionality?

कौन-सा backup medium Write-Once-Read-Many (WORM) functionality देता है?

**(A)** Hard disk **(B)** Magnetic tape
**(C)** Optical disk (CD-R / DVD-R) **(D)** Flash drive

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Optical disk (CD-R / DVD-R)** — CD-R/DVD-R optical disks are WORM — data is written once and cannot be altered, making them tamper-proof archives.

CD-R/DVD-R optical disks WORM हैं — data एक बार लिखा जाता है और बदला नहीं जा सकता, इसलिए ये tamper-proof archives हैं।

★ **Nukta:** WORM = optical write-once media (CD-R/DVD-R/BD-R). Rewritable: HDD, tape (LTO), flash, CD-RW.

WORM = optical write-once media (CD-R/DVD-R/BD-R)। Rewritable: HDD, tape (LTO), flash, CD-RW।

**Q12.** Which attack becomes possible if the SAME key is reused many times in symmetric encryption?

यदि symmetric encryption में एक ही key कई बार उपयोग हो, तो कौन-सा attack संभव हो जाता है?

**(A)** Replay attack **(B)** Known-plaintext attack
**(C)** Man-in-the-middle attack **(D)** Phishing attack

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (B) Known-plaintext attack** — With many ciphertexts under one key, an attacker can compare them with known plaintexts and deduce the key — a known-plaintext (or frequency-analysis) attack.

एक key के अंतर्गत अनेक ciphertexts होने पर attacker उन्हें known plaintexts से मिलाकर key निकाल सकता है — known-plaintext (या frequency-analysis) attack।

★ **Nukta:** One-time pads are safe ONLY with a fresh key per message. Reuse → pattern analysis. Exam favourite link: key reuse → known-plaintext.

One-time pads केवल हर message पर नई key से सुरक्षित हैं। Reuse → pattern analysis। Exam favourite link: key reuse → known-plaintext।

**Q13.** Which of the following tools is mainly used for password cracking?

निम्न में से कौन-सा tool मुख्यतः password cracking के लिए उपयोग होता है?

**(A)** Wireshark **(B)** Nmap
**(C)** Cain and Abel **(D)** Angry IP Scanner

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Cain and Abel** — Cain and Abel is a well-known password-recovery/cracking tool (sniffing, dictionary & brute-force attacks).

Cain and Abel एक प्रसिद्ध password-recovery/cracking tool है (sniffing, dictionary और brute-force attacks)।

★ **Nukta:** Security tools: Wireshark = packet analyser, Nmap = port scanner, Cain&Abel = password cracker, Angry IP Scanner = IP scanner, Snort = IDS.

Security tools: Wireshark = packet analyser, Nmap = port scanner, Cain&Abel = password cracker, Angry IP Scanner = IP scanner, Snort = IDS।

**Q14.** Which port and IP address scanner is most famous among users?

Users के बीच सबसे प्रसिद्ध port और IP address scanner कौन-सा है?

**(A)** Cain and Abel **(B)** Angry IP Scanner
**(C)** Snort **(D)** Ettercap

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (B) Angry IP Scanner** — Angry IP Scanner is the popular free tool for scanning IP ranges and open ports.

Angry IP Scanner IP ranges और open ports scan करने वाला लोकप्रिय free tool है।

★ **Nukta:** Cain&Abel = password cracking; Angry IP = IP/port scan; Snort = IDS; Ettercap = sniffing/ARP spoofing.

Cain&Abel = password cracking; Angry IP = IP/port scan; Snort = IDS; Ettercap = sniffing/ARP spoofing।

**Q15.** Match the attacks with their descriptions: (a) Phishing (b) DoS Attack (c) Man-in-the-Middle (d) Spoofing — (i) Flooding network traffic (ii) Stealing user credentials (iii) Intercepting communication (iv) Impersonation attack

हमलों को उनके विवरण से मिलाइए: (a) Phishing (b) DoS Attack (c) Man-in-the-Middle (d) Spoofing — (i) नेटवर्क ट्रैफ़िक भरना (ii) यूज़र जानकारी चुराना (iii) संचार में हस्तक्षेप (iv) पहचान की नकल

**(A)** a–ii, b–i, c–iii, d–iv **(B)** a–i, b–ii, c–iii, d–iv
**(C)** a–iii, b–i, c–ii, d–iv **(D)** a–ii, b–iii, c–i, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q3 match (codes recovered)]

✔ **Correct: (A) a–ii, b–i, c–iii, d–iv** — Phishing steals credentials (ii), DoS floods traffic (i), MitM intercepts communication (iii), Spoofing is impersonation (iv).

Phishing credentials चुराता है (ii), DoS traffic flood करता है (i), MitM संचार intercept करता है (iii), Spoofing impersonation है (iv)।

★ **Nukta:** Attack one-liners: Phishing=fake emails/sites, DoS=flood, MitM=middle interception, Spoofing=fake identity.

Attack one-liners: Phishing=fake emails/sites, DoS=flood, MitM=middle interception, Spoofing=fake identity।

## Section B — Cryptography

Plaintext/ciphertext/key, symmetric vs asymmetric ciphers, DES/AES/RSA, stream vs block, classical ciphers, digital signatures, steganography.

**Q16.** The text that gets transformed using a cipher algorithm is called:

Cipher algorithm से जिस text को बदला जाता है उसे क्या कहते हैं?

**(A)** Complex Text **(B)** Transformed Text
**(C)** Plain Text **(D)** Scalar Text

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q9]

✔ **Correct: (C) Plain Text** — The INPUT to the cipher is the plaintext; after transformation it becomes ciphertext.

Cipher का INPUT plaintext है; बदलने के बाद वह ciphertext बनता है।

★ **Nukta:** Cryptography flow: Plain text --(cipher + key)--> Ciphertext --(decipher + key)--> Plain text.

Cryptography flow: Plain text --(cipher + key)--> Ciphertext --(decipher + key)--> Plain text।

**Q17.** Which statement about plaintext, ciphertext and key is MOST accurate?

Plaintext, ciphertext और key के बारे में कौन-सा कथन सबसे सही है?

**(A)** Plaintext is generated after decryption using a public key **(B)** Ciphertext is the original readable message
**(C)** Plaintext is transformed into ciphertext using an algorithm and a key **(D)** Ciphertext and plaintext are always of the same length

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q10]

✔ **Correct: (C) Plaintext is transformed into ciphertext using an algorithm and a key** — Encryption transforms plaintext into ciphertext using a cipher algorithm together with a key — this is the precise description.

Encryption एक cipher algorithm और key के साथ plaintext को ciphertext में बदलता है — यही सटीक विवरण है।

★ **Nukta:** Remember the 3 objects: plaintext (readable), ciphertext (scrambled), key (secret parameter). Encryption = f(plaintext, key).

3 objects याद रखें: plaintext (पढ़ने योग्य), ciphertext (scrambled), key (गुप्त parameter)। Encryption = f(plaintext, key)।

**Q18.** Which of the following uses the SAME key for encryption and decryption?

निम्न में से कौन-सा encryption और decryption के लिए एक ही key उपयोग करता है?

**(A)** Symmetric key cipher **(B)** Asymmetric key cipher
**(C)** Hash function **(D)** Digital signature

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (A) Symmetric key cipher** — Symmetric cryptography uses one shared secret key for both encryption and decryption (AES, DES, Blowfish).

Symmetric cryptography encryption और decryption दोनों के लिए एक ही shared secret key उपयोग करता है (AES, DES, Blowfish)।

★ **Nukta:** Symmetric = 1 key (fast). Asymmetric = key pair (public+private, slow). Hash = one-way digest. Signature = private-sign/public-verify.

Symmetric = 1 key (तेज़)। Asymmetric = key pair (public+private, धीमा)। Hash = one-way digest। Signature = private-sign/public-verify।

**Q19.** Which of the following uses a PAIR of keys (public and private)?

निम्न में से कौन-सा keys की एक जोड़ी (public और private) उपयोग करता है?

**(A)** Symmetric cipher **(B)** Asymmetric cipher
**(C)** DES **(D)** AES

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Asymmetric cipher** — Asymmetric (public-key) cryptography uses a mathematically linked key pair — public key to encrypt, private key to decrypt (RSA, ECC, Diffie-Hellman).

Asymmetric (public-key) cryptography गणितीय रूप से जुड़ी key pair उपयोग करता है — encrypt के लिए public key, decrypt के लिए private key (RSA, ECC, Diffie-Hellman)।

★ **Nukta:** Public-key examples: RSA, Diffie-Hellman, ECC, DSA. Symmetric: DES, 3DES, AES, Blowfish, RC4.

Public-key examples: RSA, Diffie-Hellman, ECC, DSA। Symmetric: DES, 3DES, AES, Blowfish, RC4।

**Q20.** Which is FASTER in encryption and decryption?

Encryption और decryption में कौन-सा तेज़ है?

**(A)** Symmetric key cipher **(B)** Asymmetric key cipher
**(C)** Cannot decide **(D)** None of these

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (A) Symmetric key cipher** — Symmetric ciphers (AES/DES) are orders of magnitude faster than asymmetric ones (RSA) — that's why bulk data uses symmetric encryption.

Symmetric ciphers (AES/DES), asymmetric (RSA) से कई गुना तेज़ होते हैं — इसीलिए bulk data में symmetric encryption होती है।

★ **Nukta:** Speed: Symmetric >> Asymmetric. Hybrid systems (TLS/PGP) use asymmetric KEYS to exchange a symmetric SESSION key.

Speed: Symmetric >> Asymmetric। Hybrid systems (TLS/PGP) asymmetric KEYS से symmetric SESSION key का आदान-प्रदान करती हैं।

**Q21.** What is the main disadvantage of symmetric key encryption?

Symmetric key encryption की मुख्य कमी क्या है?

**(A)** It is very slow **(B)** Key distribution is difficult
**(C)** It uses two keys **(D)** It cannot encrypt text

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Key distribution is difficult** — Both sides must share the SAME secret key securely — key distribution over an insecure channel is the classic problem.

दोनों पक्षों को एक ही secret key सुरक्षित रूप से साझा करनी होती है — insecure channel पर key distribution ही classic समस्या है।

★ **Nukta:** Symmetric weakness = key distribution (n users → n(n-1)/2 keys). Asymmetric weakness = slow + key authenticity (certificates).

Symmetric कमज़ोरी = key distribution (n users → n(n-1)/2 keys)। Asymmetric कमज़ोरी = धीमा + key authenticity (certificates)।

**Q22.** Public key is used to:

Public key का उपयोग किसके लिए होता है?

**(A)** Encrypt the data **(B)** Decrypt the data

**(C)** Both encrypt and decrypt **(D)** Generate session keys only

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (A) Encrypt the data** — In public-key cryptography anyone can use the recipient's PUBLIC key to encrypt; only the recipient's PRIVATE key can decrypt.

Public-key cryptography में कोई भी प्राप्तकर्ता की PUBLIC key से encrypt कर सकता है; केवल प्राप्तकर्ता की PRIVATE key से decrypt होता है।

★ **Nukta:** Encrypt with public → decrypt with private. SIGN with private → verify with public. Exam asks both directions!

Encrypt public से → decrypt private से। SIGN private से → verify public से। Exam दोनों दिशाएँ पूछता है!

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**Q23.** In public-key cryptography, a message encrypted with the public key can only be decrypted by:

Public-key cryptography में public key से encrypt किया संदेश केवल किसके द्वारा decrypt हो सकता है?

**(A)** Another public key **(B)** The same public key

**(C)** The private key corresponding to that public key **(D)** A session key

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (C) The private key corresponding to that public key** — Only the matching private key of the key pair can decrypt what the public key encrypted.

केवल उस key pair की मेल खाती private key ही public key से encrypt किए गए संदेश को decrypt कर सकती है।

★ **Nukta:** Public/private are mathematically paired — like a lock (public) and its only key (private).

Public/private गणितीय रूप से जुड़े हैं — एक ताला (public) और उसकी एकमात्र चाबी (private) की तरह।

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**Q24.** Which statement about symmetric and asymmetric cryptography is TRUE?

Symmetric और asymmetric cryptography के बारे में कौन-सा कथन सही है?

**(A)** Symmetric cryptography provides authentication **(B)** Asymmetric cryptography is faster than symmetric

**(C)** Symmetric cryptography requires secure key exchange **(D)** Both use two different keys

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security) — Q33]

✔ **Correct: (C) Symmetric cryptography requires secure key exchange** — Symmetric cryptography requires a SECURE key exchange between the parties — that is its defining practical challenge.

Symmetric cryptography के लिए पक्षों के बीच सुरक्षित key exchange आवश्यक है — यही इसकी मुख्य व्यावहारिक चुनौती है।

★ **Nukta:** True: symmetric needs secure key exchange. False: symmetric does not inherently authenticate; asymmetric is slower; symmetric uses ONE key.

सही: symmetric को secure key exchange चाहिए। गलत: symmetric स्वयं authentication नहीं देता; asymmetric धीमा; symmetric एक key उपयोग करता है।

**Q25.** Data Encryption Standard (DES) is a:

Data Encryption Standard (DES) क्या है?

**(A)** Block cipher **(B)** Stream cipher

**(C)** Bit cipher **(D)** Byte cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (A) Block cipher** — DES encrypts data in 64-bit blocks using a 56-bit key — a classic symmetric BLOCK cipher.

DES data को 64-bit blocks में 56-bit key से encrypt करता है — एक classic symmetric BLOCK cipher।

★ **Nukta:** DES = 64-bit block, 56-bit key (symmetric). AES = 128-bit block. RC4 = stream cipher.

DES = 64-bit block, 56-bit key (symmetric)। AES = 128-bit block। RC4 = stream cipher।

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**Q26.** What is the key length of AES-256 encryption?

AES-256 encryption की key length कितनी है?

**(A)** 128 bits **(B)** 192 bits

**(C)** 256 bits **(D)** 512 bits

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (C) 256 bits** — AES-256 uses a 256-bit key (the number in the name is the key length).

AES-256 में 256-bit key उपयोग होती है (नाम में संख्या key length है)।

★ **Nukta:** AES variants: AES-128, AES-192, AES-256 (key sizes). Block size is always 128 bits.

AES variants: AES-128, AES-192, AES-256 (key sizes)। Block size हमेशा 128 bits।

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**Q27.** Which cipher method encrypts one bit or byte at a time?

कौन-सी cipher method एक बार में एक bit या byte encrypt करती है?

**(A)** Block cipher **(B)** Stream cipher

**(C)** Substitution cipher **(D)** Transposition cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Stream cipher** — A stream cipher encrypts the data one bit (or byte) at a time as a continuous stream (e.g., RC4).

Stream cipher data को एक समय में एक bit (या byte) encrypt करती है, एक निरंतर धारा के रूप में (जैसे RC4)।

★ **Nukta:** Stream = 1 bit/byte at a time (RC4, A5/1). Block = fixed-size blocks (DES/AES). Substitution/Transposition = classical techniques.

Stream = 1 bit/byte at a time (RC4, A5/1)। Block = fixed-size blocks (DES/AES)। Substitution/ Transposition = classical techniques।

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**Q28.** Which of the following statements is FALSE?

निम्न में से कौन-सा कथन गलत है?

**(A)** AES is a block cipher **(B)** RSA is a symmetric cipher

**(C)** Diffie-Hellman is used for key exchange

**(D)** DES is symmetric

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) RSA is a symmetric cipher** — RSA is an ASYMMETRIC (public-key) cipher — the statement 'RSA is symmetric' is false.

RSA एक ASYMMETRIC (public-key) cipher है — 'RSA symmetric है' कथन गलत है।

★ **Nukta:** False-statement MCQs: check each. AES=block ✓, RSA=asymmetric ✗, DH=key exchange ✓, DES=symmetric ✓.

False-statement MCQs: हर एक जाँचें। AES=block ✓, RSA=asymmetric ✗, DH=key exchange ✓, DES=symmetric ✓।

**Q29.** In the digital signature technique, when the whole message is signed using an asymmetric key, the receiver uses:

Digital signature technique में, जब पूरा संदेश asymmetric key से sign होता है, तो प्राप्तकर्ता उपयोग करता है:

**(A)** Her or his own public key **(B)** Her or his own private key

**(C)** The sender's public key **(D)** None of these

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) The sender's public key** — The sender signs with their PRIVATE key; the receiver verifies the signature using the SENDER'S PUBLIC key.

भेजने वाला अपनी PRIVATE key से sign करता है; प्राप्तकर्ता भेजने वाले की PUBLIC key से signature सत्यापित करता है।

★ **Nukta:** Signature direction: SIGN with private → VERIFY with public. Encryption direction: ENCRYPT with public → DECRYPT with private.

Signature दिशा: private से SIGN → public से VERIFY। Encryption दिशा: public से ENCRYPT → private से DECRYPT।

**Q30.** The Caesar Cipher is an example of which cryptographic technique?

Caesar Cipher किस cryptographic technique का उदाहरण है?

**(A)** Transposition cipher **(B)** Substitution cipher

**(C)** Stream cipher **(D)** Block cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Substitution cipher** — The Caesar cipher replaces each letter by another a fixed number of places down the alphabet — a classical SUBSTITUTION cipher.

Caesar cipher में हर अक्षर को वर्णमाला में कुछ स्थान आगे वाले अक्षर से बदला जाता है — एक classical SUBSTITUTION cipher।

★ **Nukta:** Classical types: Substitution (Caesar, Playfair) = replace; Transposition (Rail fence) = rearrange.

Classical types: Substitution (Caesar, Playfair) = replace; Transposition (Rail fence) = rearrange।

**Q31.** The process of detecting hidden data inside a file is known as:

किसी file के भीतर छिपे data का पता लगाने की प्रक्रिया को क्या कहते हैं?

**(A)** Cryptanalysis **(B)** Steganalysis

**(C)** Data mining **(D)** Steganography

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Steganalysis** — Steganalysis is the art of detecting/extracting data hidden by steganography.

Steganalysis, steganography से छिपाए गए data का पता लगाने/निकालने की कला है।

★ **Nukta:** Steganography = HIDING data (cover file). Steganalysis = DETECTING it. Cryptography = scrambling it.

Steganography = data छिपाना (cover file)। Steganalysis = पता लगाना। Cryptography = scramblइंग।

**Q32.** What is the output of a steganography process called?

Steganography प्रक्रिया के output को क्या कहते हैं?

**(A)** Ciphertext **(B)** Stego-object

**(C)** Encoded file **(D)** Key file

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Stego-object** — The carrier file that now contains the hidden message is called the stego-object.

वह carrier file जिसमें अब छिपा संदेश है, stego-object कहलाती है।

★ **Nukta:** Stego terms: cover file (original carrier), secret message (hidden data), stego-object (result).

Stego terms: cover file (original carrier), secret message (hidden data), stego-object (result)।

**Q33.** In steganography, the file used to hide the secret data is called:

Steganography में secret data छिपाने के लिए जिस file का उपयोग होता है उसे क्या कहते हैं?

**(A)** Secret file **(B)** Cover file

**(C)** Cipher file **(D)** Key file

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides]

✔ **Correct: (B) Cover file** — The innocent-looking carrier (image/audio/document) that hides the secret data is called the cover file.

वह मासूम दिखने वाली carrier (image/audio/document) जो secret data छिपाती है, cover file कहलाती है।

★ **Nukta:** Cover file = the disguise. Secret message = hidden data. Stego-object = cover + secret after embedding.

Cover file = छद्म वेश। Secret message = छिपा data। Stego-object = embedding के बाद cover + secret।

**Q34.** Which of the following is an example of asymmetric encryption?

निम्न में से कौन-सा asymmetric encryption का उदाहरण है?

**(A)** AES **(B)** DES

**(C)** RSA **(D)** Blowfish

[Source: SSC CGL Tier-2 repeated Qs (via Oliveboard) · also BCI_SCI ST-01]

✔ **Correct: (C) RSA** — RSA is the classic public-key (asymmetric) algorithm; AES, DES and Blowfish are all symmetric.

RSA classic public-key (asymmetric) algorithm है; AES, DES और Blowfish सब symmetric हैं।

★ **Nukta:** Asymmetric = RSA, ECC, Diffie-Hellman, DSA. Symmetric = DES, 3DES, AES, Blowfish, RC4, IDEA.

Asymmetric = RSA, ECC, Diffie-Hellman, DSA।
Symmetric = DES, 3DES, AES, Blowfish, RC4, IDEA।

## Section C — Firewalls, IPSec & Security Protocols

DMZ design, IPSec modes (transport/tunnel), SSL/TLS & HTTPS, security-protocol and mechanism layer mapping.

**Q35.** What is the advantage of setting up a DMZ (Demilitarized Zone) using a firewall?

Firewall का उपयोग करके DMZ (Demilitarized Zone) बनाने का क्या लाभ है?

**(A)** To create multiple zones of trust **(B)** Load balancing on the server

**(C)** Improving network performance **(D)** Bringing internet service into a home

[Source: RPSC Programmer 2024 Paper-1 (official key)]

✔ **Correct: (A) To create multiple zones of trust** — A DMZ creates separate trust zones — public servers (web/mail) sit in the DMZ so attackers don't directly reach the internal trusted network.

DMZ अलग trust zones बनाता है — public servers (web/mail) DMZ में रहते हैं ताकि attackers सीधे internal trusted network तक न पहुँचें।

★ **Nukta:** Official RPSC key. DMZ = buffer zone between untrusted Internet and trusted LAN (3 zones: external / DMZ / internal).

Official RPSC key। DMZ = untrusted Internet और trusted LAN के बीच buffer zone (3 zones: external / DMZ / internal)।

**Q36.** Match the security mechanisms with their OSI layers: (a) Firewall (b) Digital Signature (c) MAC Filtering (d) Port-based Filtering — (i) Network Layer (ii) Application Layer (iii) Data Link Layer (iv) Transport Layer

Security mechanisms को OSI layers से मिलाइए: (a) Firewall (b) Digital Signature (c) MAC Filtering (d) Port-based Filtering — (i) Network Layer (ii) Application Layer (iii) Data Link Layer (iv) Transport Layer

**(A)** a–i, b–ii, c–iii, d–iv **(B)** a–ii, b–i, c–iii, d–iv

**(C)** a–iv, b–ii, c–i, d–iii **(D)** a–i, b–iii, c–ii, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q4 match (codes recovered)]

✔ **Correct: (A) a–i, b–ii, c–iii, d–iv** — Firewall rules work at the network layer (IP), digital signatures are application-level, MAC filtering is data-link, and port filtering is transport-level (TCP/UDP ports).

Firewall rules network layer (IP) पर, digital signatures application level पर, MAC filtering data-link पर, और port filtering transport level (TCP/UDP ports) पर काम करते हैं।

★ **Nukta:** Mechanism-layers: Firewall L3/L4 (IP), Port filter L4, MAC filter L2, Digital signature L7. Packet-filtering firewall = L3/L4.

Mechanism-layers: Firewall L3/L4 (IP), Port filter L4, MAC filter L2, Digital signature L7। Packet-filtering firewall = L3/L4।

**Q37.** Match the security protocols with their OSI layers: (a) IPSec (b) SSL/TLS (c) PGP (d) WPA2 — (i) Application Layer (ii) Network Layer (iii) Transport/Presentation Layer (iv) Data Link Layer (Wi-Fi)

Security protocols को OSI layers से मिलाइए: (a) IPSec (b) SSL/TLS (c) PGP (d) WPA2 — (i) Application Layer (ii) Network Layer (iii) Transport/Presentation Layer (iv) Data Link Layer (Wi-Fi)

**(A)** a–ii, b–iii, c–i, d–iv **(B)** a–ii, b–i, c–iii, d–iv

**(C)** a–iii, b–ii, c–i, d–iv **(D)** a–i, b–ii, c–iii, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q1 match (codes recovered)]

✔ **Correct: (A) a–ii, b–iii, c–i, d–iv** — IPSec secures at the network layer (L3), SSL/TLS at transport/presentation, PGP is an application-layer mail security tool, and WPA2 protects Wi-Fi at the data-link layer.

IPSec network layer (L3) पर, SSL/TLS transport/presentation पर, PGP application-layer mail security tool है, और WPA2 Wi-Fi को data-link layer पर सुरक्षित करता है।

★ **Nukta:** Protocol layers: IPSec L3, SSL/TLS L4-7 (exam: transport/presentation), PGP L7, WPA2 L2, Kerberos L7.

Protocol layers: IPSec L3, SSL/TLS L4-7 (exam: transport/presentation), PGP L7, WPA2 L2, Kerberos L7।

**Q38.** IPSec is MOST commonly used to implement:

IPSec मुख्यतः किसके लिए उपयोग होता है?

**(A)** Secure email **(B)** Secure web browsing

**(C)** Virtual Private Network (VPN) **(D)** Wireless security

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Virtual Private Network (VPN)** — IPSec provides encrypted tunnels over the internet — the foundation of most site-to-site and remote-access VPNs.

IPSec internet पर encrypted tunnels देता है — अधिकांश site-to-site और remote-access VPNs की नींव।

★ **Nukta:** IPSec = VPN tunneling (ESP/AH). Secure email = PGP/SMIME. Secure web = SSL/TLS. Wireless = WPA2.

IPSec = VPN tunneling (ESP/AH)। Secure email = PGP/SMIME। Secure web = SSL/TLS। Wireless = WPA2।

**Q39.** IPSec operates in which mode(s)?

IPSec किस/किन mode(s) में काम करता है?

**(A)** Transport mode only **(B)** Tunnel mode only

**(C)** Both transport mode and tunnel mode **(D)** Application mode

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Both transport mode and tunnel mode** — IPSec has two modes — transport mode (protects the payload, end-to-end) and tunnel mode (protects the whole IP packet, gateway-to-gateway).

IPSec के दो modes हैं — transport mode (payload सुरक्षित, end-to-end) और tunnel mode (पूरा IP packet सुरक्षित, gateway-to-gateway)।

★ **Nukta:** IPSec modes: Transport (payload only, host-to-host) & Tunnel (full packet, gateway-to-gateway/VPN).

IPSec modes: Transport (केवल payload, host-to-host) और Tunnel (पूरा packet, gateway-to-gateway/VPN)।

**Q40.** In IPSec Tunnel mode:

IPSec Tunnel mode में:

**(A)** Only the payload is encrypted **(B)** Only the header is encrypted

**(C)** The entire IP packet is encrypted **(D)** No encryption is used

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) The entire IP packet is encrypted** — Tunnel mode encrypts the ENTIRE original IP packet (header + payload) and adds a new header — used for VPN gateways.

Tunnel mode पूरे original IP packet (header + payload) को encrypt करके नया header जोड़ता है — VPN gateways के लिए उपयोग।

★ **Nukta:** Tunnel = whole packet + new header (gateway VPN). Transport = payload only (host-to-host).

Tunnel = पूरा packet + नया header (gateway VPN)। Transport = केवल payload (host-to-host)।

**Q41.** Which mode of IPSec provides end-to-end security between two hosts?

IPSec का कौन-सा mode दो hosts के बीच end-to-end security देता है?

**(A)** Transport mode **(B)** Tunnel mode

**(C)** Static mode **(D)** Gateway mode

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (A) Transport mode** — Transport mode secures communication directly between two hosts (end-to-end) — only the payload is protected.

Transport mode दो hosts के बीच सीधे संचार सुरक्षित करता है (end-to-end) — केवल payload सुरक्षित होता है।

★ **Nukta:** End-to-end (host↔host) = Transport mode. Gateway↔gateway (VPN) = Tunnel mode.

End-to-end (host↔host) = Transport mode। Gateway↔gateway (VPN) = Tunnel mode।

**Q42.** The SSL/TLS handshake protocol is mainly used for:

SSL/TLS handshake protocol मुख्यतः किसके लिए उपयोग होता है?

**(A)** Data encryption **(B)** Data compression

**(C)** Key exchange and authentication **(D)** Packet routing

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Key exchange and authentication** — The handshake negotiates cipher suites, authenticates the server (certificates) and exchanges keys for the session.

Handshake cipher suites का निर्णय करता है, server को authenticate करता है (certificates) और session के लिए keys का आदान-प्रदान करता है।

★ **Nukta:** TLS parts: Handshake = keys+authentication; Record = encryption+integrity. HTTPS = HTTP over TLS (port 443).

TLS parts: Handshake = keys+authentication; Record = encryption+integrity। HTTPS = HTTP over TLS (port 443)।

**Q43.** Which of the following is used in HTTPS for encryption?

HTTPS में encryption के लिए निम्न में से क्या उपयोग होता है?

**(A)** SSL/TLS **(B)** FTP

**(C)** ARP **(D)** NAT

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (A) SSL/TLS** — HTTPS = HTTP over SSL/TLS — the TLS layer encrypts the HTTP traffic (port 443).

HTTPS = HTTP over SSL/TLS — TLS layer HTTP traffic को encrypt करता है (port 443)।

★ **Nukta:** HTTPS = HTTP + SSL/TLS (443). HTTP = 80 (no encryption). FTP = 21. ARP = IP→MAC. NAT = address translation.

HTTPS = HTTP + SSL/TLS (443)। HTTP = 80 (encryption नहीं)। FTP = 21। ARP = IP→MAC। NAT = address translation।

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**Q44.** In which layer of the network architecture is the Secure Socket Layer (SSL) used?

Secure Socket Layer (SSL) network architecture की किस layer में उपयोग होती है?

**(A)** Physical layer **(B)** Session layer

**(C)** Application layer **(D)** Presentation layer

[Source: Unit Test-4 Solution (OCR), p6 (bank key)]

✔ **Correct: (B) Session layer** — Exam banks place SSL at the Session layer — it establishes a secure session between application and transport (technically it sits between L4 and L7).

Exam banks SSL को Session layer पर रखते हैं — यह application और transport के बीच secure session बनाती है (तकनीकी रूप से यह L4 और L7 के बीच है)।

★ **Nukta:** Exam convention: SSL = Session layer (secure session setup + negotiation). Note: TLS = successor of SSL.

Exam convention: SSL = Session layer (secure session setup + negotiation)। नोट: TLS, SSL का उत्तराधिकारी है।

## ★ Nukte Rapid-Fire — Exam Pointers (quick revision)

One-line question (left) with its exam answer + trick (right). Revise these before the exam.

| | |
|---|---|
| CIA triad?<br>CIA triad? | ✔ Confidentiality, Integrity, Availability.<br>Confidentiality, Integrity, Availability। |
| Goal of each: interception / modification / DoS / spoofing?<br>हर एक का लक्ष्य: interception / modification / DoS / spoofing? | ✔ Confidentiality / Integrity / Availability / Authentication.<br>Confidentiality / Integrity / Availability / Authentication। |
| Active vs passive attack?<br>Active बनाम passive attack? | ✔ Active = changes/disrupts data. Passive = only observes (eavesdrop, traffic analysis).<br>Active = data बदलता/बाधित करता है। Passive = केवल देखता है (eavesdrop, traffic analysis)। |
| Virus phases?<br>Virus phases? | ✔ Dormant → Propagation → Triggering → Execution (payload runs).<br>Dormant → Propagation → Triggering → Execution (payload चलता है)। |
| WORM medium?<br>WORM medium? | ✔ Optical disk (CD-R/DVD-R) — write once, read many.<br>Optical disk (CD-R/DVD-R) — एक बार लिखें, कई बार पढ़ें। |
| Symmetric vs asymmetric keys?<br>Symmetric बनाम asymmetric keys? | ✔ 1 shared key (fast, key-distribution problem) vs key pair public+private (slow).<br>1 shared key (तेज़, key-distribution समस्या) बनाम key pair public+private (धीमा)। |
| Encrypt/decrypt vs sign/verify direction?<br>Encrypt/decrypt बनाम sign/verify दिशा? | ✔ Encrypt public → decrypt private. Sign private → verify public.<br>Encrypt public → decrypt private। Sign private → verify public। |
| DES / AES / RSA basics?<br>DES / AES / RSA basics? | ✔ DES = 64-bit block/56-bit key. AES = 128-bit block (128/192/256 keys). RSA = asymmetric.<br>DES = 64-bit block/56-bit key। AES = 128-bit block (128/192/256 keys)। RSA = asymmetric। |

| | |
|---|---|
| Stream vs block cipher?<br>Stream बनाम block cipher? | ✔ Stream = 1 bit/byte at a time (RC4). Block = fixed blocks (DES/AES).<br>Stream = 1 bit/byte at a time (RC4)। Block = fixed blocks (DES/AES)। |
| Caesar cipher type?<br>Caesar cipher प्रकार? | ✔ Classical substitution (replace letters by shifting).<br>Classical substitution (अक्षरों को खिसका कर बदलना)। |
| Steganography terms?<br>Steganography terms? | ✔ Cover file (carrier) + secret message → stego-object. Detection = steganalysis.<br>Cover file (carrier) + secret message → stego-object। पहचान = steganalysis। |
| False-friend: RSA is symmetric?<br>False-friend: RSA symmetric है? | ✔ NO — RSA is ASYMMETRIC. AES/DES/Blowfish are symmetric.<br>नहीं — RSA ASYMMETRIC है। AES/DES/Blowfish symmetric हैं। |
| Security tools?<br>Security tools? | ✔ Wireshark = sniff, Nmap = port scan, Cain&Abel = password crack, Angry IP = IP scan, Snort = IDS.<br>Wireshark = sniff, Nmap = port scan, Cain&Abel = password crack, Angry IP = IP scan, Snort = IDS। |
| Key reuse risk?<br>Key reuse जोखिम? | ✔ Known-plaintext / frequency analysis attacks become possible.<br>Known-plaintext / frequency analysis attacks संभव हो जाते हैं। |
| DMZ?<br>DMZ? | ✔ Buffer zone between internet & trusted LAN — public servers face outside, internals stay safe.<br>Internet और trusted LAN के बीच buffer zone — public servers बाहर, internals सुरक्षित। |
| IPSec modes?<br>IPSec modes? | ✔ Transport = payload only (host-to-host end-to-end). Tunnel = whole packet (gateway VPN).<br>Transport = केवल payload (host-to-host end-to-end)। Tunnel = पूरा packet (gateway VPN)। |

| | |
|---|---|
| IPSec / SSL / PGP / WPA2 layers?<br>IPSec / SSL / PGP / WPA2 layers? | ✔ Network / Transport-Presentation / Application / Data Link (Wi-Fi).<br>Network / Transport-Presentation / Application / Data Link (Wi-Fi)। |
| Firewall / port filter / MAC filter / signature layers?<br>Firewall / port filter / MAC filter / signature layers? | ✔ Network / Transport / Data Link / Application.<br>Network / Transport / Data Link / Application। |
| SSL/TLS handshake does?<br>SSL/TLS handshake करता है? | ✔ Key exchange + server authentication (certificates). Record protocol does encryption.<br>Key exchange + server authentication (certificates)। Encryption Record protocol करता है। |
| HTTPS = ?<br>HTTPS = ? | ✔ HTTP + SSL/TLS on port 443.<br>HTTP + SSL/TLS port 443 पर। |
| DoS vs DDoS?<br>DoS बनाम DDoS? | ✔ Single-source flooding vs distributed (many zombies/botnet) flooding.<br>Single-source flooding बनाम distributed (कई zombies/ botnet) flooding। |
| Attack examples map?<br>Attack examples map? | ✔ Phishing=steal credentials; MitM=intercept; Spoofing=impersonate; Smurf=DoS variant.<br>Phishing=credentials चुराना; MitM=intercept; Spoofing=impersonate; Smurf=DoS variant। |